

Auteur: Tiffany Kegels
Studierichting: Informatica
Oefeningenreeks 1B nr. 43.9

$$2z^2 - 7z - 4 = i(2z + 1)$$

$$\Rightarrow 2z^2 + (-7 - 2i)z + (-4 - i) = 0$$

$$D = (-7 - 2i)^2 - 4 \cdot 2 \cdot (-4 - i) = 77 + 36i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = 77 \\ 2xy = 36 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 77 \\ xy = 18 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 77 \\ x^2(-y)^2 = 324 \end{cases}$$

$$\lambda^2 - 77\lambda - 324 = 0$$

$$\Delta = (-77)^2 - 4 \cdot (-324) = 7225 = 85^2$$

$$\lambda_1 = \frac{77 + 85}{2} = 81$$

$$\lambda_2 = \frac{77 - 85}{2} = 4$$

$$k = (9 + 2i)$$

$$z_1 = \frac{(7 + 2i) + (9 + 2i)}{4} = \frac{16 + 4i}{4} = 4 + i$$

$$z_2 = \frac{(7 + 2i) - (9 + 2i)}{4} = \frac{-2}{4} = \frac{-1}{2}$$

References